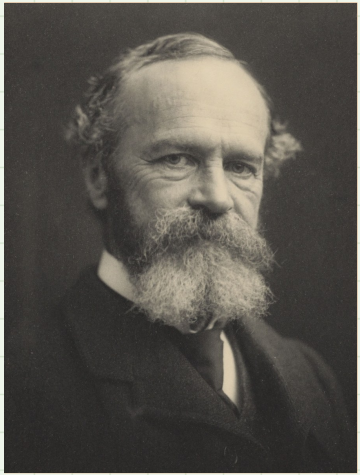
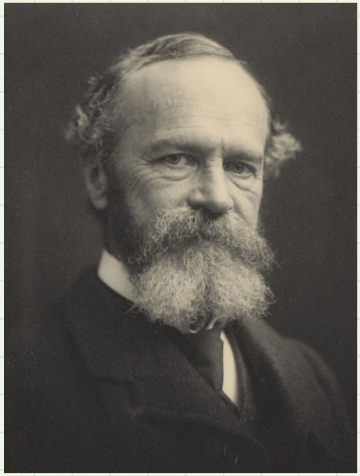


Notman Studios, 1903
William James

William James

1842 to 1910

Trained as a physician, he taught psychology and philosophy at Harvard



Notman Studios, 1903
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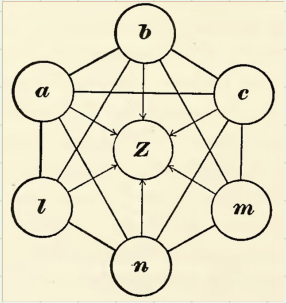
William James

1842 to 1910

Trained as a physician, he taught psychology and philosophy at Harvard

The Principles of Psychology, 1890:
two volumes, twelve years in the writing

In 1890 William James published *The Principles of Psychology*, which starts the study of the mind from the nervous system. Its third chapter collects what physiologists knew about when a nerve centre discharges.

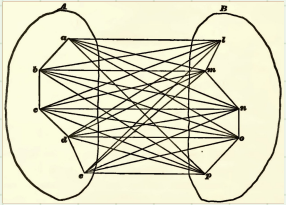


William James, 1890
Fig. 44: tensions that rouse Z

The law is this, that a stimulus which would be inadequate by itself to excite a nerve-centre to effective discharge may, by acting with one or more other stimuli (equally ineffectual by themselves alone) bring the discharge about. The natural way to consider this is as a summation of tensions which at last overcome a resistance. The first of them produce a 'latent excitement' [. . .]; the last is the straw which breaks the camel's back.

William James, *The Principles of Psychology* (1890), vol. 1, p. 82.

Why does one thought call up another? James's chapter on association answers with the brain: points of the cortex that were active together tend to excite each other again.



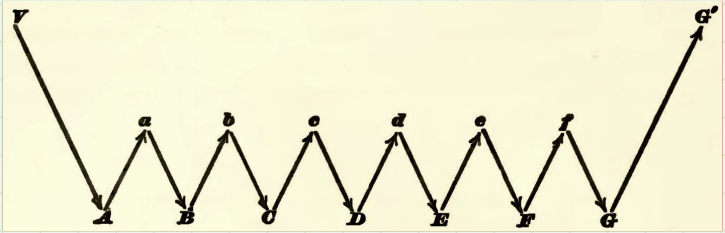
William James, 1890
Fig. 40: every point of A into every point of B

The amount of activity at any given point in the brain-cortex is the sum of the tendencies of all other points to discharge into it, such tendencies being proportionate (1) to the number of times the excitement of each other point may have accompanied that of the point in question; (2) to the intensity of such excitements; [. . .]

William James, *The Principles of Psychology* (1890), vol. 1, p. 567.

In the chapter on habit James follows a series of movements, A, B, C and so on. While we learn it, the mind checks each step; as a habit, it runs by itself once started.

[. . .] no sooner has the conscious thought or volition instigated movement A*, than* A*, through the sensation* a *of its own occurrence, awakens* B *reflexly;* B *then excites* C *through* b*, and so on till the chain is ended, when the intellect generally takes cognizance of the final result.*



William James, 1890
Fig. 24: the chain of a habit

William James, *The Principles of Psychology* (1890), vol. 1, p. 116.

Each passage of a current deepens its path

Why does practice make an act easier? Earlier in the same chapter, James guesses at a physical cause in the paths that nerve currents take through the brain.

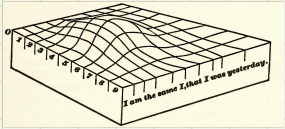


Meindert Hobbema, 1689
The Avenue at Middelharnis

A path once traversed by a nerve-current might be expected to follow the law of most of the paths we know, and to be scooped out and made more permeable than before; and this ought to be repeated with each new passage of the current. Whatever obstructions may have kept it at first from being a path should then, little by little, and more and more, be swept out of the way, until at last it might become a natural drainage-channel.

William James, *The Principles of Psychology* (1890), vol. 1, pp. 107 to 108.

James tests his law of association on two lines of Tennyson's *Locksley Hall* that share the words *the ages*. Recited from memory, each line goes on with its own next word.



William James, 1890
Fig. 33: a thought, word by word

Take, to fix our ideas, the two verses from 'Locksley Hall': "I, the heir of all the ages *in the foremost files of time," and "For I doubt not through* the ages *one increasing purpose runs." Why is it that when we recite from memory one of these lines, and get as far as* the ages*, that portion of the* other *line which follows [. . .] does not also sprout out of our memory [. . .]? Simply because the word that follows* the ages *has its brain-process awakened not simply by the brain-process of* the ages *alone, but by it* plus *the brain-processes of all the words preceding* the ages.

William James, *The Principles of Psychology* (1890), vol. 1, pp. 567 to 568.



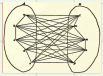
Tensions add up until they overcome a resistance.

The hinge



Tensions add up until they overcome a resistance.

The hinge



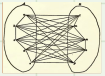
Each point sums what all the others send it.

Hinge-fan networks



Tensions add up until they overcome a resistance.

The hinge



Each point sums what all the others send it.

Hinge-fan networks

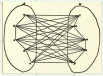


Each link wakes the next, to the end of the chain.

Hinge-chain networks



Tensions add up until they overcome a resistance. — The hinge



Each point sums what all the others send it. — Hinge-fan networks



Each link wakes the next, to the end of the chain. — Hinge-chain networks



Each passage deepens the path a little more. — Training by gradient descent



Tensions add up until they overcome a resistance.

The hinge



Each point sums what all the others send it.

Hinge-fan networks



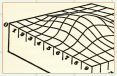
Each link wakes the next, to the end of the chain.

Hinge-chain networks



Each passage deepens the path a little more.

Training by gradient descent



The next word is called up by all the words before it.

Language models